



Secular coefficients mini-course

Talk 4

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Recall Keating and Snaith gave the prediction

$$M_k(T) := \frac{1}{T} \int_0^T \left| \zeta \left(\frac{1}{2} + it \right) \right|^{2k} dt \sim a_k g_k (\log T)^{k^2}$$

$$\text{for } g_k = \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}.$$

This was motivated by an analogous result for $U(N)$:

Theorem 1 (Keating–Snaith)

We have

$$\mathbb{E} |\det(I - U_N)|^{2k} = \prod_{j=1}^N \frac{\Gamma(j) \Gamma(j+2k)}{(\Gamma(j+k))^2} \sim g_k N^{k^2}.$$

How can one approach moments of ζ ? One uses the approximation

$$\zeta(1/2 + it) \approx \sum_{n \leq T} n^{-1/2 - it}$$

which is quite good when $T \asymp t$.

Then observe that the functions $(n^{-it})_{n \leq T}$ are essentially orthogonal in $L^2([0, T])$, so

$$\begin{aligned} \frac{1}{T} \int_0^T |\zeta(1/2 + it)|^2 dt &\approx \frac{1}{T} \int_0^T \left| \sum_{n \leq T} n^{-1/2 - it} \right|^2 dt \\ &\approx \frac{1}{T} \int_0^T \sum_{n \leq T} |n^{-1/2 - it}|^2 dt = \sum_{n \leq T} \frac{1}{n} \sim \log T. \end{aligned}$$

Conrey–Gamburd looked at the following limit, which they called a pseudomoment:

$$PM_k(x) := \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left| \sum_{n \leq x} \frac{1}{n^{1/2+it}} \right|^{2k} dt.$$

Because the integrand only depends on p^{it} for $p \leq x$, and $(p^{it})_{p \leq x} \xrightarrow[T \rightarrow \infty]{d} (\alpha(p))_{p \leq x}$ when $t \sim \text{Uniform}([0, T])$, then

$$PM_k(x) = \mathbb{E} \left| \sum_{n \leq x} \frac{\alpha(n)}{n^{1/2}} \right|^{2k},$$

i.e. moments of Steinhaus.

By expanding the $2k$ th power, and using $\mathbb{E}[\alpha(n)\overline{\alpha(m)}] = \delta_{n,m}$, we obtain

$$PM_k(x) = \sum_{\substack{n_1, \dots, n_k \leq x \\ m_1, \dots, m_k \leq x \\ \prod n_i = \prod m_j}} \frac{1}{\sqrt{n_1 \cdots n_k}} \frac{1}{\sqrt{m_1 \cdots m_k}}.$$

Theorem 2 (Conrey–Gamburd, 2003)

We have

$$PM_k(x) \sim a_k g'_k (\log x)^{k^2}$$

for

$$g'_k = \text{vol}(\mathcal{P}_k), \quad \mathcal{P}_k = \{(x_{i,j}) \in \mathbb{R}_{\geq 0}^{k^2} : \sum_{i=1}^k x_{i,j} \leq 1, \sum_{j=1}^k x_{i,j} \leq 1\}.$$

The proof of Conrey–Gamburd is complex-analytic: define

$$F(s_1, \dots, s_k, t_1, \dots, t_k) = \sum_{\substack{n_1, \dots, n_k \\ m_1, \dots, m_k \geq 1 \\ \prod n_i = \prod m_j}} \prod_{i=1}^k n_i^{-s_i} \prod_{j=1}^k m_j^{-t_j}$$

and apply Perron's formula to each of the variables of F .

Motivation for Conrey–Gamburd (I)

The expression

$$\sum_{n \leq x} \frac{\alpha(n)}{\sqrt{n}}$$

is analogous to the following sum in finite fields:

$$\sum_{\deg f \leq n} \frac{\alpha_q(f)}{|f|^{1/2}},$$

where α_q is Steinhaus on polynomials. This is the sum of the first n coefficients of $\mathcal{L}(u/\sqrt{q}, \alpha_q)$, so it is also analogous to

$$\sum_{i \leq n} c_i.$$

Motivation for Conrey–Gamburd (II)

Diaconis–Gamburd (2003):

$$\mathbb{E} \left| \sum_{n \leq N} c_n \right|^{2k}$$

is the number of $k \times k$ arrays of nonnegative integers, where each row sum and each column sum does not exceed N ; this grows like $g'_k N^{k^2}$.

Is there a more transparent link between the two results? Yes!

Suppose we are given positive integers $(R_{i,j})_{1 \leq i,j \leq k}$. Define

$$n_i = \prod_{j=1}^k R_{i,j}, \quad m_j = \prod_{i=1}^k R_{i,j}.$$

Then $\prod_{i=1}^k n_i = \prod_{j=1}^k m_j = \prod_{1 \leq i,j \leq k} R_{i,j}$, and morally

$$\sum_{\substack{n_1, \dots, n_k \leq x \\ m_1, \dots, m_k \leq x \\ \prod n_i = \prod m_j}} \frac{1}{\sqrt{n_1 \cdots n_k}} \frac{1}{\sqrt{m_1 \cdots m_k}} \approx \sum_{\substack{\prod_{j=1}^k R_{i,j} \leq x \\ \prod_{i=1}^k R_{i,j} \leq x}} \frac{1}{\prod_{1 \leq i,j \leq k} R_{i,j}}.$$

These conditions are multiplicative analogues of the arrays counted in Diaconis–Gamburd.

Issue: the map that takes a $k \times k$ array $(R_{i,j})_{i,j \leq k}$ to the pair of tuples $(\prod_{j=1}^k R_{i,j})_{i=1}^k$ and $(\prod_{i=1}^k R_{i,j})_{j=1}^k$ is not injective.

Define

$$\text{Co}_k = \{R \in M_k(\mathbb{N}) : \gcd(R_{p,q}, R_{r,s}) = 1 \text{ when } p < r \text{ and } q > s\}.$$

A given element is coprime to elements above and to the right of it, as well as to those below and to the left of it.

Lemma 1 (Vaughan–Wooley, Montgomery–Soundararajan)

$$(R_{i,j})_{i,j \leq k} \mapsto ((\prod_{j=1}^k R_{i,j})_{i=1}^k, (\prod_{i=1}^k R_{i,j})_{j=1}^k)$$

is a bijection from Co_k onto the set of pairs of tuples $(n_1, \dots, n_k)$ and $(m_1, \dots, m_k)$ with $\prod_{i=1}^k n_i = \prod_{j=1}^k m_j$.

This leads to the following rigorous equality:

$$\sum_{\substack{n_1, \dots, n_k \leq x \\ m_1, \dots, m_k \leq x \\ \prod n_i = \prod m_j}} \frac{1}{\sqrt{n_1 \cdots n_k}} \frac{1}{\sqrt{m_1 \cdots m_k}} = \sum_{\substack{\prod_{j=1}^k R_{i,j} \leq x \\ \prod_{i=1}^k R_{i,j} \leq x \\ (R_{i,j})_{1 \leq i,j \leq k} \in \text{Co}_k}} \frac{1}{\prod_{1 \leq i,j \leq k} R_{i,j}}.$$

How to analyze? Write as

$$\sum_{(R_{i,j})_{1 \leq i,j \leq k} \in \text{Co}_k} \frac{1}{\prod_{1 \leq i,j \leq k} R_{i,j}} f \left(\left(\frac{\log R_{i,j}}{\log x} \right)_{i,j} \right)$$

where f is the indicator of the polytope $\mathcal{P}_k$ that appeared before.

Tauberian trick: if one can analyze this for $f(\vec{s}) = e^{-\vec{s} \cdot \vec{\lambda}}$, one gets for free an estimate for indicators of convex bounded sets.

The general formula has the shape

$$\sum_{(R_{i,j})_{1 \leq i,j \leq k} \in \mathcal{C}o_k} \frac{1}{\prod_{1 \leq i,j \leq k} R_{i,j}} f \left(\left(\frac{\log R_{i,j}}{\log x} \right)_{i,j} \right) \sim a_k (\log x)^{k^2} \int_{\mathbb{R}^{k^2}} f$$

for

$$a_k = \prod_p a_k(p), \quad a_k(p) := (1 - 1/p)^{k^2} \sum_{i=0}^{\infty} \binom{i+k-1}{k-1}^2 p^{-i}.$$

Lemma of V–W/M–S: one may reduce to the case where $(R_{i,j})_{i,j}$ are powers of a fixed prime, say 2:

$$R_{i,j} = 2^{r_{i,j}}.$$

Define

$$\text{Co}_k^+ = \{r \in M_k(\mathbb{N}_0) : \min(r_{p,q}, r_{r,s}) = 0 \text{ when } p < r \text{ and } q > s\}.$$

The lemma reduces to constructing a bijection

$$(r_{i,j})_{i,j \leq k} \mapsto \left(\left(\sum_{j=1}^k r_{i,j} \right)_{i=1}^k, \left(\sum_{i=1}^k r_{i,j} \right)_{j=1}^k \right)$$

from Co_k^+ onto the set of pairs of tuples $(n_1, \dots, n_k)$ and $(m_1, \dots, m_k)$ with $\sum_{i=1}^k n_i = \sum_{j=1}^k m_j$.

To construct the inverse of the bijection we use the greedy algorithm. Known as the North West Corner Method.

The North West Corner Method

1	$m_1=4$	$m_2=6$	$m_3=2$
$n_1=5$	4	0	0
$n_2=3$	0	0	0
$n_3=4$	0	0	0

2	$m_1=4$	$m_2=6$	$m_3=2$
$n_1=5$	4 → 1	0	0
$n_2=3$	0	0	0
$n_3=4$	0	0	0

3	$m_1=4$	$m_2=6$	$m_3=2$
$n_1=5$	4	1 ↓	0
$n_2=3$	0	3	0
$n_3=4$	0	0	0

4	$m_1=4$	$m_2=6$	$m_3=2$
$n_1=5$	4	1	0
$n_2=3$	0	3 ↓	0
$n_3=4$	0	2	0

5	$m_1=4$	$m_2=6$	$m_3=2$
$n_1=5$	4	1	0
$n_2=3$	0	3	0
$n_3=4$	0	2 → 2	

Moments of $\sum_{n \leq N} c_n$ are related to arrays where row and column sums are bounded by N .

What about moments of $\sum_{n=0}^N (-1)^n \text{Sc}_n(U_N)$?

It is natural to ask, because

$$\sum_{n=0}^N (-1)^n \text{Sc}_n(U_N) = \det(I - U_N)$$

is analogous to

$$\zeta(1/2 + it).$$

Given a $k \times k$ array A of nonnegative integers, a South–East chain is a sequence of distinct entries of A in which each entry is located weakly to the right of and weakly below the preceding entry.

The length of a chain is the sum of entries in the chain.

We define

$$\text{SE}(A)$$

as the maximal length of a South–East chain in A .

The support of an array obtained in the North West Corner Method is a South–East chain.

Theorem 3 (Baik–Rains, 2001)

The expectation

$$\mathbb{E}\left[\prod_{i=1}^k \text{Sc}_{n_i}(U_N) \prod_{j=1}^k \overline{\text{Sc}_{m_j}(U_N)}\right]$$

equals the number of $k \times k$ arrays A of nonnegative integers, where the i th row sums to n_i , the j th column sums to m_j , and additionally

$$\text{SE}(A) \leq N.$$

Consequently,

$$\mathbb{E}|\det(I - U_N)|^{2k}$$

equals the number of $k \times k$ arrays A of nonnegative integers, where the row sums and column sums are bounded by N , and additionally $\text{SE}(A) \leq N$.

When studying c_N , we encountered arrays with row-sum and column-sum constraints.

When studying $\alpha(n)$, we encountered arrays with row-product and column-product constraints.

This suggests defining the *multiplicative* length of a chain is the product of the entries in the chain, and

$$\mathrm{SE}^\times(R)$$

as the maximal *multiplicative* length of a South–East chain, and

The analogy of U_N and ζ suggests that

$$\frac{1}{T} \int_0^T \left| \zeta \left(\frac{1}{2} + it \right) \right|^{2k} \sim \sum_{\substack{R_{i,j} \in \text{Co}_k \\ \text{SE}^\times(R) \leq T}} \frac{1}{\prod_{1 \leq i,j \leq k} R_{i,j}}.$$

We prove this is consistent (i.e. equivalent) with the Keating–Snaith prediction.

Theorem 4 (Gorodetsky–Rodgers)

$$\sum_{\substack{R_{i,j} \in \text{Co}_k \\ \text{SE}^\times(R) \leq T}} \frac{1}{\prod_{1 \leq i,j \leq k} R_{i,j}} \sim a_k g_k (\log T)^{k^2}.$$

Moreover, $g_k = \text{vol}(\mathcal{Q}_k)$ where

$$\mathcal{Q}_k = \{(x_{i,j}) \in \mathbb{R}_{\geq 0}^{k^2} : \sum_{i=1}^k x_{i,j} \leq 1, \sum_{j=1}^k x_{i,j} \leq 1, \text{SE}(x) \leq 1\}.$$

To explain Baik and Rainss proof we need symmetric function theory.

Schur polynomials $s_\lambda(x_1, \dots, x_N)$ are indexed by partitions. They are important in the study of U_N because when x_i are replaced with eigenvalues of U_N , we obtain irreducible characters of U_N , and in particular we obtain:

Fact 1

$$\mathbb{E}[s_\lambda(U_N) \overline{s_\mu(U_N)}] = \delta_{\lambda, \mu} \mathbf{1}_{\ell(\lambda) \leq N}.$$

There are many bases of symmetric polynomials, and we know how to pass between them.

Example: $\text{Tr}(U_N^k)$ is the k -power sum symmetric polynomial in the eigenvalues of U_N , denoted p_k . By expressing p_k and its powers as a linear combination of Schur polynomials, Diaconis and Shahshahani proved that

$$\mathbb{E}|\text{Tr}(U_N^k)|^{2m} = k^m m!$$

holds for $mk \leq N$, hence $\text{Tr}(U_N^k)/\sqrt{k} \xrightarrow[N \rightarrow \infty]{d} N_{\mathbb{C}}(0, 1)$ when $k = o(N)$.

$\text{Sc}_n(U_N)$ is the n th elementary symmetric polynomial in the eigenvalues of U_N , denoted $e_n(U_N)$. We have

$$\prod_{i=1}^k \text{Sc}_{n_i}(U_N) = e_{\mathbf{n}}(U_N)$$

where $\mathbf{n} = (n_1, \dots, n_k)$ and $e_{\mathbf{n}} := \prod_i e_{n_i}$.

Fact 2

If $|\lambda| = n$ then

$$e_\lambda = \sum_{\mu \vdash n} K_{\mu', \lambda} s_\mu.$$

where the sum is over partitions of n , μ' is the conjugate partition, and K is the Kostka number which counts *semistandard Young tableaux of shape μ' and weight λ* .

Thus, if $\sum_{i=1}^k n_i = \sum_{j=1}^k m_j = n$ then

$$\mathbb{E}\left[\prod_{i=1}^k \text{Sc}_{n_i}(U_N) \prod_{j=1}^k \overline{\text{Sc}_{m_j}(U_N)}\right] = \sum_{\substack{\mu \vdash n \\ \ell(\mu) \leq N}} K_{\mu', \mathbf{n}} K_{\mu', \mathbf{m}} = \sum_{\substack{\mu \vdash n \\ \mu_1 \leq N}} K_{\mu, \mathbf{n}} K_{\mu, \mathbf{m}}.$$

Fact 3

The RSK algorithm gives a bijection between

- Pairs (P, Q) of semistandard Young tableaux of equal shape with entries in $\{1, 2, \dots, k\}$
- $k \times k$ arrays A of nonnegative integers.

It has the following properties:

- The common shape of P and Q has size which is equal to the sum of entries in A .
- The weight of Q (resp. P) is given by the row (resp. column) sums of A .
- The longest SE-chain of A is equal to the largest part in the common shape.

Matrix A

2	1	1	4
1	1	0	2 row sums
0	1	0	$\mu = (4, 2, 1)$
3	3	1	1 (a partition)
column sums			
$\nu = (3, 3, 1)$			(a partition)

Repeat the pair (i, j) exactly a_{ij} times.

Two-line array

1	1	1	1	2	2	3
1	1	2	3	1	2	2

sorted by top row, then by bottom row

A longest increasing subsequence
in the bottom row is
 $1, 1, 1, 2, 2,$
so its length is 5.

Final SSYT's

P and Q have common shape $\lambda = (5, 2)$.

P					Q				
1	1	1	2	2	1	1	1	1	3
2	3				2	2			

Longest part of the shape: $\lambda_1 = 5$

content(P)
= $\nu = (3, 3, 1)$
= column sums
of A

content(Q)
= $\mu = (4, 2, 1)$
= row sums
of A

Thank you for listening.